SpectraPlan

Initial spectroscopy assessment

Analysis date: 2026-10-18 · Round 2

Enquiry

We manufacture tablets from a powder blend containing an active pharmaceutical ingredient and excipients. Laboratory checks tell us whether the blend is on target, but results arrive after the material has moved through the press. We want to assess whether a fiber based measurement in the feed frame could follow API concentration changes. Our first pilot should compare reasonable measurement options before we commit to production equipment.

Customer clarifications

- powder blend: What is the target API concentration range and the required measurement error tolerance for the powder blend? Answer: For this synthetic development check, use retained process samples for an offline pilot. Quantitative acceptance limits are still undecided.

Assessment at a glance

Provisional

powder blend (stream_1); material: powder blend containing an active pharmaceutical ingredient and excipients; phase: powder; requested outcomes: follow API concentration changes; reported reference checks: Laboratory checks tell us whether the blend is on target

Application and conditional method hypothesis are supported by evidence for in-line blend potency monitoring in a tablet press feed frame using fiber-based NIR and Raman techniques.

Customer requirements

- powder blend: Label: powder blend (customer statement: powder blend)
- powder blend: Measurement goals: follow API concentration changes (customer statement: follow API concentration changes)
- powder blend: Sample matrix: powder blend containing an active pharmaceutical ingredient and excipients (customer statement: powder blend containing an active pharmaceutical ingredient and excipients)
- powder blend: Sample phase: powder (customer statement: powder blend containing an active pharmaceutical ingredient and excipients)
- powder blend: Reference assays: Laboratory checks tell us whether the blend is on target (customer statement: Laboratory checks tell us whether the blend is on target)

Information still needed

- What is the target API concentration range and the required measurement error tolerance for the powder blend?

- Are there specific optical or physical properties of the powder blend that affect fiber-based measurement?
- Specific optical or physical properties of the powder blend that affect fiber-based measurement
- Details of the current laboratory reference assay (e.g., HPLC, UV-Vis) and its uncertainty

Methods worth investigating

NIR · powder blend

Conditional on clarification

Source excerpt (application_example): A multispectral NIRaman fiber probe has industrial grade design that combines NIR reflection and Raman optical techniques in a compact shaft that fits in a tablet press feed frame. [S01 p.15]

Measurement geometry: diffuse reflection probe. Measurement location: feed frame of a rotary tablet press.

Spectral range (Undetermined): Not established

Coverage interval is undetermined for the proposed configuration; usable equipment coverage remains unverified. [S01 p.5]

Assumptions to confirm

- The powder blend is representative of the scattering media and powder sample types listed for NIR diffuse reflection.
- The feed frame geometry allows for the insertion of a compact fiber probe.

Method limitations

- Source excerpt (general_method): method_no: 2; method: NIR spectroscopy; measurement_technique: diffuse reflection probe; physical_phenomenon: diffuse reflection; sample_type: scattering media, powder; optical_path_or_depth: 0.5-1 mm; sensitivity: average; selectivity: average; cost: average [T01 method_comparison row 9 (S01 Table 1 p.6)]
- Source excerpt (general_method): Measurement of the weak Raman effect requires special probes but is recorded in the visible and NIR spectral regions. [S01 p.5]

Raman · powder blend

Conditional on clarification

Source excerpt (application_example): A multispectral NIRaman fiber probe has industrial grade design that combines NIR reflection and Raman optical techniques in a compact shaft that fits in a tablet press feed frame. [S01 p.15]

Measurement geometry: Raman probe. Measurement location: feed frame of a rotary tablet press.

Spectral range (Undetermined): Not established

Coverage interval is undetermined for the proposed configuration; usable equipment coverage remains unverified. [S01 p.5]

Assumptions to confirm

- The powder blend is representative of the liquid, powder, and solid sample types listed for Raman spectroscopy.
- The feed frame geometry allows for the insertion of a compact fiber probe.

Method limitations

- Source excerpt (general_method): method_no: 3; method: Raman spectroscopy; measurement_technique: Raman probe; physical_phenomenon: Raman scattering (emission); sample_type: liquid, powder, solid; optical_path_or_depth: 2-5 mm; sensitivity: average; selectivity: high; cost: high [T01 method_comparison row 13 (S01 Table 1 p.6)]
- Source excerpt (general_method): Measurement of the weak Raman effect requires special probes but is recorded in the visible and NIR spectral regions. [S01 p.5]

Risks and limitations

- powder blend: Source excerpt (general_method): method_no: 2; method: NIR spectroscopy; measurement_technique: diffuse reflection probe; physical_phenomenon: diffuse reflection; sample_type: scattering media, powder; optical_path_or_depth: 0.5-1 mm; sensitivity: average; selectivity: average; cost: average [T01 method_comparison row 9 (S01 Table 1 p.6)]
- powder blend: Source excerpt (general_method): method_no: 3; method: Raman spectroscopy; measurement_technique: Raman probe; physical_phenomenon: Raman scattering (emission); sample_type: liquid, powder, solid; optical_path_or_depth: 2-5 mm; sensitivity: average; selectivity: high; cost: high [T01 method_comparison row 13 (S01 Table 1 p.6)]

Questions before proceeding

- powder blend: What is the target API concentration range and the required measurement error tolerance for the powder blend? Why this matters: Customer clarification is needed before finalizing the proposed feasibility study.
- powder blend: What is the acceptable response time for the in-line measurement to be useful for process control decisions? Why this matters: Customer clarification is needed before finalizing the proposed feasibility study.
- powder blend: What are the specific optical or physical properties of the powder blend (e.g., particle size, moisture content, color) that might affect fiber-based measurement? Why this matters: Customer clarification is needed before finalizing the proposed feasibility study.
- powder blend: What are the details of the current laboratory reference assay (e.g., HPLC, UV-Vis) used to determine if the blend is on target, including its uncertainty and turnaround time? Why this matters: Customer clarification is needed before finalizing the proposed feasibility study.

Proposed feasibility study

powder blend

Proposed: Assess whether fiber-based NIR and Raman measurements in the feed frame can follow API concentration changes in the powder blend, comparing reasonable measurement options before committing to production equipment.

Samples

- Proposed: Retained process samples from the powder blend

- Proposed: Representative acceptable and off-spec material states

Measurements and reference assays

- Proposed: NIR diffuse reflection spectra
- Proposed: Raman spectra
- Proposed: pair spectra and independent laboratory references at aligned sample locations and times; confirm assay identity, uncertainty and sampling lag with the customer.
- Proposed: stream_1: agree quality variables and independently characterize representative acceptable/ off-specification material before analyte calibration or exact configuration selection.

Comparisons

- Proposed (stream_1): evaluate NIR individually against aligned independent laboratory reference measurements.
- Proposed (stream_1): evaluate Raman individually against aligned independent laboratory reference measurements.
- Proposed (stream_1): retain a combined model only if it improves the client-defined decision on independent material over the individual methods.

Variation to cover

- Proposed: API concentration changes
- Proposed: Excipient variation (proposed)

Validation approach

- Proposed (stream_1): keep related scans together and hold out entire independent run units for evaluation.
- powder blend: Characterization; independent unit: Run

Acceptance criteria

- powder blend: Required error: Agree with customer (Unknown)
- powder blend: Response time raw: Agree with customer (Unknown)
- powder blend: Bias: Agree with customer (Unknown)
- powder blend: Robustness: Agree with customer (Unknown)

Go / no-go decision

- Proposed: Go/no-go proposal: proceed only if evaluation on independent material meets agreed customer-defined error, bias, response time and robustness criteria. Missing thresholds remain open; inconclusive results require clarification or further characterization.

Next steps

- Target API concentration range

- Required measurement error tolerance
- Acceptable response time
- Specific optical/physical properties of the powder blend
- Details of the current laboratory reference assay
- stream_1: clarify the target and independent characterization/reference method.
- stream_1: agree required_error acceptance threshold with the customer.
- stream_1: agree response_time_raw acceptance threshold with the customer.
- stream_1: agree bias acceptance threshold with the customer.
- stream_1: agree robustness acceptance threshold with the customer.

Evidence and references

S01 p.15

Data processing for pharmaceutical tasks relies on "low-level data fusion," where the synchronized NIR and Raman spectra are merged into a single augmented matrix. Then regression models, such as Partial Least Squares (PLS), are applied to these fused datasets. By combining Raman and NIR, these models can predict API potency or coating thickness with higher accuracy and robustness than single-method models.

Fig. 6. A multispectral NIRaman fiber probe has industrial grade design that combines NIR reflection and Raman optical techniques in a compact shaft that fits in a tablet press feed frame.

T01 method_comparison row 9 (S01 Table 1 p.6)

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Source note: Source: "Multispectral Laboratory of Optical Analysis Methods", Table 1, page 6 (file S01).

Source note: Transcribed from the rendered page because PDF text extraction interleaves the columns.

Source note: "mu" in the original means micrometres; written here as "um".

Source note: Merged cells of the original are expanded, so each row is independent.

Source note: sensitivity / selectivity / cost are the original qualitative ratings (low / average / high), not measured values.

Source note: "ATR probe*" carries an asterisk in the original; no footnote is printed on that page.

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S01 p.5

For example, NIR spectroscopy, with its relatively large penetration depth (up to several millimetres) even into dense solid samples due to low absorption and strong scattering, allows the use of ordinary glass for optics, as well as standard lamps as light sources, making it ideal for online applications, including remote measurements via fiber optic probes. In contrast, strong absorption in the mid-IR region limits the corresponding spectral method to the analysis of very thin samples (usually less than 50 µm), but it opens up the possibility of using attenuated total reflectance (ATR) by connecting appropriate attachments and probes.

Measurement of the weak Raman effect requires special probes but is recorded in the visible and NIR spectral regions. Moreover, the spectra of vibrational transitions in Raman and mid-IR spectroscopy carry complementary qualitative and quantitative information about the sample's composition.
UV-visible spectroscopy is based on electronic transitions between molecular energy levels. It has very high sensitivity, especially in the UV region (190-360 nm), but requires the use of special light sources and fiber optic probes. However, the strong overlap of spectral signals in UV spectra complicates the analysis of real samples and makes the use of chemometric algorithms a necessary component of qualitative mixture analysis methods. Another highly sensitive method, UV/Vis fluorimetry, is based on the secondary emission of excited molecules of certain substances, most often of biological origin. However, most chemicals do not exhibit this property.
Therefore, fluorimetry is primarily used in medicine, biology, and biotechnology.
Spectra from different methods vary greatly, both in their nature and in the information they contain. Mid-IR band spectra contain well-defined peaks (absorption bands), while NIR, UV, and fluorescence spectra contain broad, highly overlapping signals, forming a so-called continuous spectrum.
Extracting the information necessary for quantitative and qualitative analysis from optical spectra requires special methods for analyzing multidimensional data, called the above-mentioned term chemometrics. Chemometric algorithms are implemented in specialized software that enables preliminary data preparation (preprocessing), construction, validation, and application of predictive models, specifically calibration or classification models.
Immersible fiber optic probes operating in various spectral regions from UV to mid-IR are increasingly being used for measuring liquid and soft or powdered solid samples. They can be used to obtain absorption, diffuse reflectance or transmission, Raman, and ATR spectra. The use of these probes can be essential for in-situ analysis—in the field or in an ongoing process. Online measurements require high-speed spectral acquisition.
Often, analytical results must be obtained in real time to be used for management decisions.
An analysis of individual spectral methods, their capabilities, and limitations, demonstrates that developing an optimal instrumental methodology for solving a given analytical problem requires considering many different factors. Thus, selecting a spectral method for a specific analytical problem is a complex task requiring specialized knowledge in chemistry, physics, and mathematics. This is especially true for equipping an entire laboratory for future needs, since the need for a particular optical analysis method is difficult to predict in advance. In the equipment package offered here, called a multispectral laboratory, experts from various fields have already worked hard to select the optimal solutions.

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This is an initial assessment and a proposed study, not an experimentally validated performance claim. Confirm suitability and acceptance criteria through testing.